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Course Code: MLA0305

Course Name: REINFORCEMENT LEARNING

Proposed Algorithm for TD (0)-Based Recommendation System

Step 1: Define Recommendation Environment

Let

- S= $\{s_1, s_2, \ldots, s_n\}$: Set of user states
- A= $\{a_1, a_2, \ldots, a_m\}$: Set of recommendation actions
- R: Reward function
- $(V(S))$:State-value function
- (α): Learning rate
- (γ): Discount factor

Step 2: Observe Current User State

Current user state is defined as

$$S_t = s_i \tag{1}$$

Eq. (1) represents the current interaction state of the user, such as browsing history or previously selected items. This state serves as the input for generating the next recommendation.

Step 3: Select Recommendation

The recommendation policy selects an action using

$$A_t = \pi(S_t) \tag{2}$$

Eq. (2) determines the recommendation action according to the current policy. The selected item is presented to the user for interaction.

Step 4: Compute Reward

The reward is obtained from user feedback.

$$R_t = f(\text{Click, Purchase, Rating}) \tag{3}$$

Eq. (3) computes the immediate reward based on the user's response. Positive interactions increase the reward, whereas negative feedback reduces it.

Step 5: Update State Value

The TD (0) update rule is

$$V(S_t) = V(S_t) + \alpha[R_t + \gamma V(S_{t+1}) - V(S_t)] \tag{4}$$

Eq. (4) updates the estimated state value using the temporal-difference error. The update combines the immediate reward with the estimated value of the next state.

Step 6: Repeat Until Convergence

The stopping condition is

$$|V_{\text{new}} - V_{\text{old}}| < \varepsilon \tag{5}$$

Eq. (5) indicates that learning has converged when successive updates become smaller than a predefined threshold.

Proposed Algorithm for SARSA-Based Recommendation System

Step 1: Initialize Recommendation Parameters

Let

- (S): User states
- (A): Recommendation actions
- (Q(S,A)): Action-value function
- (\alpha): Learning rate
- (\gamma): Discount factor

Step 2: Observe Current State

$$S_t = s_i \tag{6}$$

Eq. (6) represents the current user state obtained from interaction history and contextual information.

Step 3: Select Recommendation Action

$$A_t = \pi(S_t) \tag{7}$$

Eq. (7) selects a recommendation using the current policy, balancing exploration and exploitation.

Step 4: Observe Next State

$$S_{t+1} \tag{8}$$

Eq. (8) denotes the new user state after interacting with the recommended item.

Step 5: Select Next Recommendation

$$A_{t+1} = \pi(S_{t+1}) \tag{9}$$

Eq. (9) chooses the next action according to the same behaviour policy used during learning.

Step 6: Update Q-Value

$$Q(S_t, A_t) = Q(S_t, A_t) + \alpha[R_t + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)] \tag{10}$$

Eq. (10) updates the action-value estimate using the reward received and the Q-value of the next state-action pair actually selected by the policy.

Proposed Algorithm for Q-Learning-Based Recommendation System

Step 1: Define Recommendation Environment

Let

- (S): User states
- (A): Recommendation actions
- (Q(S,A)): Action-value function
- (R): Reward

Step 2: Observe Current User State

$$S_{t}=s_{i} \tag{11}$$

Eq. (11) represents the current state describing the user's browsing or interaction context.

Step 3: Choose Recommendation

$$A_t = \pi(S_t) \tag{12}$$

Eq. (12) selects a recommendation action based on the exploration–exploitation policy.

Step 4: Receive Reward

$$R_t = f(\text{Click, Purchase, Rating}) \tag{13}$$

Eq. (13) computes the reward from user interactions, reflecting the quality of the recommendation.

Step 5: Update Q-Value

$$Q(S_t, A_t) = Q(S_t, A_t) + \alpha \left[R_t + \gamma \max Q(S_{t+1}, a) - Q(S_t, A_t)\right] \tag{14}$$

a

Eq. (14) updates the action-value function using the maximum estimated future reward, enabling the agent to learn the optimal recommendation strategy.

Step 6: Improve Recommendation Policy

$$\pi^*(S) = \arg\max_a Q(S, a) \qquad \textbf{(15)}$$

Eq. (15) selects the recommendation with the highest learned Q-value, resulting in the optimal policy for future interactions.

This format matches the style in your reference images:

- **Step heading**
- **LaTeX label**
- **One-row, two-column borderless table** (formula | equation number) •
  Equation explanation below each table